OVERCHARGE // KIOTD STUDIOS

Level 1 Roadmap

How we finish Neon Rise and make it a fully fleshed-out, dynamic first level

> **The goal.** A first level that teaches the whole absorb, carry, spend loop by letting the player experiment, that rewards looking around, that has life in the streets, and that feels good to touch: every jump, absorb and gate has feedback. Levels 2 to 6 stay paused until this one is done.

> **The plan in one line.** Rebuild the rules that Level 1 needs, give Chief the Builder tools to place them, make the art and the story, build the level vertically, then tune it by playing it.

Prepared by Kiro, Technical Director, for Chief
Updated October 16, 2026
Sources: docs/GAME_DESIGN_V2.md, docs/LEVEL_DESIGN_V2.md, the orders CODEX 01, AKI 26 and ORCHA 19, and KIRO_STATUS.md

1. Where we are today

100 x 18	~40 s	3 NPCs	2779 / 7
Level 1 size today (tiles)	until the first spend, today	two still say "New line"	tests passing / failing (none from new work)

Level 1 today is a flat street with three streetlamps (4 charge each), an exit gate that costs 8, one checkpoint, a pit, a high-and-low fork, and about 129 decorations. The top 10 rows of the screen are empty. It teaches the loop, but the first spend happens at the exit about 40 seconds in, and the last 1,000 px before the exit is a flat, quiet walk. Nothing makes a sound except footsteps.

Done so far

Item	Status	What exists
Game Design V2 and Level Design V2	Done	Every decision tagged CONFIRMED, BUILT, RECOMMENDED or OPEN. Older design documents archived unchanged.
Pip and gate behavior documented and tested	Done	53 tests describing how pips and the 8-charge gate interact.
Stale provers fixed	Done	Level 2 checks now pass for real. Found that Level 5's fence can be jumped over.
Juice module (first pass)	Built	Dust, spark streams, gate burst, shake, hit-stop. Off switch and reduced-motion support. Not yet judged by Chief in play.
Audio bus (first pass)	Built	Synthesised sounds and absorb/charge hums. Hum-into-Level-2 bug fixed. Not yet heard by Chief.
Objective card wording	Done	Stage 1 reads "Try absorbing power from a street lamp".
Charge system settled	Decided	Bar 10, pip 10, one pip slot at the start (20 held), each Capacitor Module adds a slot. Q: a pip pays the gate's 8 and 2 returns to the bar. Exits stay 8. Charge starts slow and speeds up as the bar fills.
Story canon	Approved	Malik Carter, Meridian Industries, MIOR, Project Tempest, six chapters (Chief and Orcha).
Codex phases A and B	Built	Q pip gates, two-way charged objects, charge speed (to be replaced), duck, ability controls, exit reveal.
Aki: Builder tools and art	On her branch	Charge readout, platform path tool, reactive NPC fields, ladder badge, level-height fixes (committed and pushed, not yet on the live line). Four art pieces failed Chief's review and went to Codex.
Relay pylon in Level 1	Placed	Chief liked it. The hidden room preview was scrapped by Chief and removed; only the pylon stays. Its job: the start of a lit cable line to the exit (cable comes with the vertical build).

2. What "finished" means

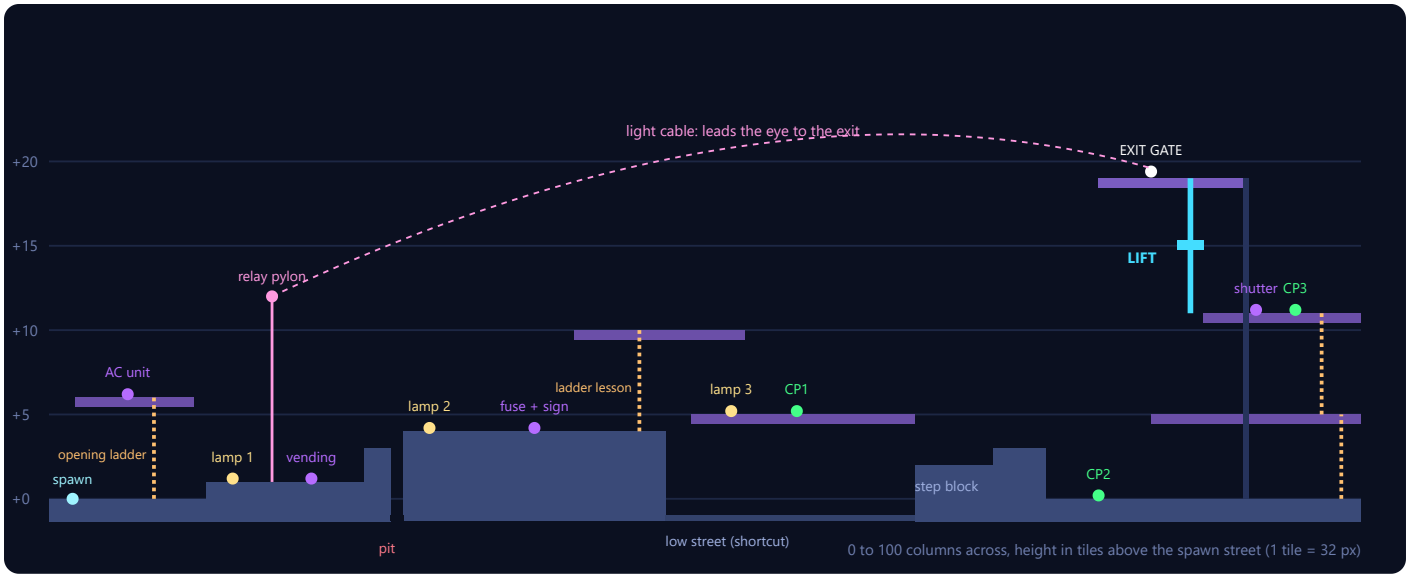
Level 1 is done when every line below is true. Each one has a way to check it, so nobody has to argue about it.

#	The test	How we check it

1	A new player absorbs, carries and spends within 60 seconds of spawning, with no instructions.	Timed run. The model says first absorb at 8 s, first spend at 12 s.
2	The main route takes 90 to 120 seconds. Exploring can take longer.	Timed run. The model says about 110 s; sprinting past everything is about 36 s.
3	No stretch of the route longer than 8 seconds without something to see, hear or do.	Script scans the finished level file.
4	The exit is shown early, and the gate says "Charge gate to open." above it.	Look at the first screen; look at the gate.
5	The fork is a real choice and both branches finish the level from zero charge.	Reachability prover walks both with real physics.
6	Every charged object can be reached again from the gate by two-way paths (ladder, stairs, lift), never only by a one-way drop.	Prover check.
7	The level's own charge total pays the exit (8) plus every required spend. Chief sets the total.	Builder readout plus a test reading the level file.
8	No placeholder text. Hint lines are observations, never instructions.	Chief reads every line.
9	Every action has feedback: jump, land, absorb, charge, gate, checkpoint, hurt. Sound can be switched off and so can motion.	Chief plays it; tests cover the switches.
10	Test suites pass except failures that are named and explained. Chief signs off in the Builder.	Suite report; Chief's play-through.

3. The level we are building

The street you built stays as the ground floor. We add 16 rows on top (18 to 34 rows) and build a tower at the east end. The player climbs ladders, rides a charge-powered lift, and reaches the exit gate on the gatehouse roof. This is a draft; Chief checks it in the Builder before anything is final.



The route

- **Plaza.** Spawn, a pedestrian, the relay pylon and light cable that point up and east to the exit, and an optional ladder.

The tenant sequence (replaces the scrapped hidden room)

- **Walkway.** Lamp 1 (first absorb), then the vending machine (first spend, with a visible payoff).
- **Pillar, pit, rooftop.** The jump, lamp 2, the smoke-break workers, the fuse box and sign.
- **Ladder lesson.** A short, safe climb to the upper roof, taught alone.
- **Fork.** High strip (lamp 3, the streetlamp woman, checkpoint) or the low street shortcut.
- **Tower.** Two ladder floors, drone graffiti, the shutter, then the lift to the gate.

A new four-panel apartment building near the plaza has a tenant in a window who is too hot. A small light-switch plate sits on the wall at player height. The player pays 2 charge there, the AC unit under her window turns on, she is relieved, and an achievement is awarded automatically. Optional; the three lamps still pay the exit.

That gives the player a reason to turn around without making the level run backwards.

What the game must learn first

Charged objects work both ways. The lift moves up and down along a path the author draws. A banked pip opens a gate instantly on **Q** (it pays the 8 and gives 2 back). Charging starts slow when the bar is empty and speeds up as it fills. Ducking becomes a standard move.

4. Who does what

Owner	Order	Job	Output
Codex	CODEX 03, then 02	First redo the art (window NPC, a NEW four-panel tenant apartment that matches the rooftop tiles, pip pickup, Capacitor Module to match Chief's concept #3, small wall-plate switch) and the engine pieces (plate switch, tenant reacts to the AC, AC turns on, achievements system). Then the 10-unit bar with pip slots, the new charge speed and the charge table. Phase C still open.	Art, code, tests, a report per task
Aki	AKI 27	Get her Builder tools onto the live line (rebase and push), read the plan, then build the vertical Level 1 as a separate draft file for Chief to compare and swap. The failed art is no longer hers.	Builder features and a Level 1 draft
Orcha	ORCHA 19	The story canon is approved. Next: a short story bible that points at it, and every line the player reads in Level 1. Writing only.	Documents; no code
Kiro	this plan	Design documents, orders and review; prover and route checks; juice and audio; hero duck and double-jump frames; movement retune; Level 1 music.	Code, tests, art checks, reports
Chief	the Builder	Builds the level, picks the story premise and names, tunes the feel in play, approves art and gives final sign-off.	Level files; decisions

Rule for everyone. Chief owns the level files. Agents build tools, art and rules; only Chief places things in Level 1. Orders are committed files, work goes to your own branch, nothing merges to the live line without review.

5. The phases

Status, October 16. Phase 0 is done. Phase 1 is mostly done: Codex built the energy rules, Aki committed the Builder tools, the story is canon. Phase 2 has started: the failed art went to Codex, Aki builds the vertical Level 1 draft, and a first version of the AC secret and the relay pylon are already in Level 1 for Chief to see. The charge system was redesigned on October 15 (bar 10, pip 10, pip slots), so the energy tasks are being redone as CODEX 02.

There are no calendar dates here on purpose. The order is set by what depends on what, and each phase ends with a gate that must pass before the next starts.

Phase 0: Foundations — DONE

Design documents, decision ledger, prover fixes, first juice and audio, orders written.
Gate passed: docs link-checked, 2719 tests passing with 5 explained failures.

Phase 1: Rules and tools — Codex A, Aki 1.x, Orcha step 1

- Codex: two-way charged objects, the uncharged field, Q pip use, charge-speed scaling, chest pip pickup.
- Aki: the Uncharged checkbox, level-total readout, platform path tool, row adding, ladder test level.
- Orcha: three story premises, then wait for Chief's pick.
- Kiro: reviews each report; keeps the test baseline.

Gate: an object can be placed uncharged in the Builder, charged in play, absorbed back, and survive a checkpoint rewind; charge is conserved.

Phase 2: Movement, world and art — Codex B and C, Aki art, Kiro, Orcha steps 2 and 3

- Codex: duck and drop-through, per-level abilities, fail-state detection, path-based platforms (the lift), gate label, objectives, reactive NPC lines, events for juice.
- Aki: pylon, light cable, beacon, AC recolour, pip pickup (each approved by Chief first). Codex now owns the apartment, window NPC and wall switch art.
- Kiro: hero duck and double-jump frames, movement retune with before-and-after numbers, re-check every route in all six levels.
- Orcha: story bible, then every Level 1 line.

Gate: a rider stays on a rising and falling platform; the duck dodges a laser and leaves existing routes intact; art passes pixel checks and Chief's eye.

Phase 3: Build the level — Chief in the Builder

- Add rows, place ladders, the lift and its path, charged objects, NPCs with their lines, the pylon and cable, the tenant sequence.
- Kiro re-runs the timing, the 8-second rule, the fork and the reachability checks on the real file.

Gate: the level can be finished from zero charge by both fork branches; every charged object is reachable by two-way paths.

Phase 4: Juice and polish — Kiro, with Chief playing

- Tune landing, absorb and gate feedback; add ladder, lift and duck effects; reactions for each charged object (vending machine hums, sign lights, shutter rises, AC fan spins).
- Ambience under the rain, then Level 1 music. Every effect keeps its off switch.
- Tune the charge-speed curve and lift speed by playing.

Gate: Chief plays the first minute and each moment gets a reaction he likes, with and without motion effects.

Phase 5: Verify and ship	Kiro, then Chief

Timed runs (with someone who has not seen the level), full test suite, route re-check across all six levels, `level_guard`, Builder module version, then Chief's sign-off and merge to the live line.

6. The feel plan: juice by moment

The first pass is built. The rest comes after the mechanics exist.

Moment	Feedback	State
Jump and land	Dust puffs, small stretch, shake that scales with fall height (only on long drops), jump and land sounds.	**Built**
Absorb	Spark stream from the lamp to the hero, lamp flicker and dim, dry-flash and power-down thump when it empties, a hum whose pitch rises with the transfer rate.	**Built** (hum, sparks); lamp flicker steps **next**
Charge and spend	Spark stream hero to object; each charged object reacts: vending machine hums and drops a can, fuse box lights its sign, shutter rises, AC fan spins.	Stream built; reactions **need Phase 1 rules**
Gate opens	Burst, short freeze, shake, rising sweep and thump, then the level lights pulse.	Burst, freeze, sound **built**; light pulse **next**
Checkpoint, chest, hurt	Colour bursts and a sound each.	**Built**
Ladder, lift, duck	Rung sounds, rising hum with rail sparks, a clunk on stopping; duck and double-jump poses and sounds.	**After Codex B and C**
The street	Steam, birds, window figures, passing cars, puddle ripples, thunder, lamps that stay dark.	**Phase 4**
Camera and UI	Look-ahead camera, checkpoint flash, level-complete banner building from the gate glow.	Look-ahead needs the retune; **Phase 4**
Music	A Level 1 track and a layered ambience bed.	**Phase 4 (Kiro)**

> **Safety built in.** `?juice=off` removes the visuals, `?audio=off` removes the sound, and the browser's reduced-motion setting removes shake and freeze. Shake is capped at 6 px, the freeze at 0.12 s, particles at 400. None of it can change a gameplay result.

7. Risks, and what we do about them

Risk	Why it matters	What we do
PixelLab art comes out weak	Earlier output had white borders; Orcha's results were poor.	Pixel checks on every piece, contact sheets, Chief approves before use, stop after two bad tries. Hard animation (hero frames) stays with Kiro, who built the hero pipeline.
Four pieces of art failed Chief's review	Apartment too flat and too blue, the window NPC's animations read wrongly, the pip pickup is a blob, the Capacitor Module is not the concept's design.	Codex redoes them with strict checks and a written spec; Chief approves each before anything is registered or placed. Kiro measured the apartment: 95% blue-dominant against 43 to 86% for the real families, and flat.

Three people editing the same files	`player.js`, `main.js` and the import map in the HTML files conflict easily.	Codex owns `player.js`. Kiro only listens through events and keeps the `main.js` hooks small. Import maps are regenerated after merging, never merged by hand.
Ladders have never been used in a real level	The engine has them and tests, but no shipped level uses one.	Build the opening ladder first in a test copy and climb it.
The lift is new engine work	Rider carry on two axes can hover, drop through, or crush.	Codex C1 specifies and tests rise, fall, loop, drained mid-ride and rewind.
Nobody has heard the sound or felt the effects	Tests prove structure, not taste.	Chief plays the first minute and says what to change.
No story yet	NPC lines, the secret clue and the drone foreshadow depend on it.	Orcha writes; placeholders stay until the premise is chosen.
Level 5's fence can be jumped over	Found by the new prover; Level 5 is paused.	Reported only. Not fixed until Levels 2 to 6 resume.
Codex's scope was assumed from `AGENTS.md`	Kiro has not seen his recent work.	Chief confirms Codex is the right owner before he starts.

8. What we need from Chief

1. Save your Capacitor Module concept sheet into the repo at `docs/reference/capacity_upgrade_concept_sheet.png`. Kiro cannot copy a chat attachment, and Codex needs it to match item #3.
2. Look at the relay pylon in Level 1 and say whether it should lead a lit cable to the exit. Answer two small questions on the tenant sequence: switch only, and is 2 the right price?
3. Approve or fail each redone art piece as Codex delivers it. Decide on the AC recolour's full-bleed border (Aki's waiver request).
4. Review Aki's vertical Level 1 draft when she delivers it, and swap it in if you like it.
5. Play the first minute with and without `?juice=off` and say what to change in the effects and sounds. Check that the charge hum stops at the gate.
6. Tune in play: the charge-speed floor, the lift speed, and the charge table numbers.

9. How we keep track

- **One source of truth.** `docs/GAME_DESIGN_V2.md` and `docs/LEVEL_DESIGN_V2.md` hold every decision with a tag. If a document and the code disagree, the V2 documents win until Chief says otherwise.
- **Orders are committed files.** Each agent reports per phase with commit, files, tests and what they did not do, then stops for review.
- **Test baseline.** 2719 passing, 5 failing (the Level 5 fence, the "New line" placeholders and two suites that already crash). A new failure is a stop sign.
- **Status.** `KIRO_STATUS.md` gets an entry after every change, ending with the commit and push line.

Roadmap prepared by Kiro for Chief, updated October 16, 2026. This is a plan, not a promise of dates: the phases finish when their gates pass.